

Buzz Buzz Buzz Goes the Honeybee

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Buzz Buzz Buzz Goes the Honeybee How Honeybees Use Sound & Vibration to Communicate Inside the hive, there is no light, no spoken language - but it is not silent. The air hums. The comb trembles. Every buzz, every vibration, means something. Honeybees live in a world where sound is felt as much as it is heard - where communication travels not through words, but through waves of vibration pulsing through their bodies and their home. Honeybees use this powerful, vibrational language to navigate, coordinate, and thrive in the dark symphony of the hive. A beekeeper benefits from understanding honeybee sound and vibration because these signals reveal the colony's needs, health, foraging activity, and swarming behavior, enabling better hive management and safer, more productive beekeeping. First, we'll get on the same page about what sound actually is. Then we'll look at how bees use vibrations as a language of their own. We'll see how those vibrations travel through the hive, how other bees pick them up, and the special sensors - called chordotonal organs - that make that possible. I'll touch on the role of thoracic muscle contractions in creating these signals, and finally, we'll put it all together with the most famous example: the waggle dance. What is sound? All sound is a form of vibration that travels through a medium - air, water, solid - capable of being detected by a hearing organ. An example of an airborne vibration is a guitar string being plucked, creating pressure waves in air. An example of vibration through a solid - called a substrate-borne vibration - is a knock on the door. If you place your ear on the door, you're sensing vibrations through the door, but the vibrations of the door create pressure waves in air that your ears pick up when not pressed against the door. Honeybees do not intentionally make sounds by picking up or manipulating objects. Their acoustic and vibrational communication relies primarily on wing buzzing, body vibrations, and movements transmitted through the comb or air. Honeybees create "sound" using vibrations - creating pressure waves with wingbeats or thoracic muscle contractions - their "flight motor." I want to briefly share some of the honeybee dances but we'll take a closer look at the waggle dance in particular later on. 1. The Waggle Dance Let's start with the most famous: the waggle dance. This one's all about direction and distance - it's how foragers tell the hive where to find food, water, or even a new nest site, if it's more than about 100 meters away. The dance starts with a straight run, where the bee waggles her abdomen and produces short thoracic pulses - these are rhythmic vibrations coming from her flight muscles. Then, she loops around, alternating sides, creating a figure-eight pattern. What's amazing is that this dance literally maps the landscape: The angle of the waggle run, relative to vertical, tells the direction of the resource relative to the sun. And the duration of the waggle phase? That tells you how far it is - the longer the waggle, the farther the source. This groundbreaking behavior was first decoded by Karl von Frisch, who later won the Nobel Prize for this work in 1967. The dancers are foragers, and how long they keep dancing

depends on a few things: the quality of the food, the distance, whether other bees give a stop signal, and the overall needs of the hive. 2. The Round Dance Now, let's zoom in – literally. When the food source is very close to the hive, within about 50 to 100 meters, bees use what's called the round dance. In this one, the forager simply runs in a circle, switching directions every so often. There's no directional information – it's more of a shout that says: "Hey! There's food close by. Go out and look!" Karl von Frisch initially thought this was a completely separate dance from the waggle. Later on, Tom Seeley studied it in more detail. In "Piping Hot Bees and Boisterous Buzz Runners," he pointed out that the line between waggle and round dances is blurrier than we thought – especially when bees are visiting feeders around 20 meters away. The followers of a round dance exit the hive and search for food nearby, without specific location information other than, "Go outside the entrance and take a look around." 3. The Tremble Dance Next up is the tremble dance – and this one's not about finding food at all. It's about processing it. Here's the situation: a forager returns with nectar but finds a traffic jam. Not enough bees are unloading. So, she starts to tremble. This dance looks totally different from the waggle – she walks slowly through the hive, trembling and making brief contact with other bees along the way. The goal? To recruit more receiver bees – to get that nectar unloaded faster. Originally, von Frisch thought the tremble dance was just a variation of the waggle. But in the 1990s, Seeley and others showed it's a completely separate behavior – it's not about navigation, but about regulating labor inside the hive. What's causing the tremble? The bee's thoracic flight muscles, the same ones used in flight. But here, the wings usually aren't flapping. Instead, those muscle contractions create shudders that travel through the legs into the comb, sending a tactile signal to others nearby. 4. The Shaking Signal (Also called the Jerking or Vibration Dance) This one is short, intense, and very clear in its message: "Get moving." The shaking signal is a general-purpose arousal signal, used to increase activity in the hive – especially to ramp up foraging or food processing. Here's how it works: one bee grabs another and vibrates her body rapidly for just 1 to 2 seconds. It doesn't carry any information about location. It's more like a wake-up call – especially common early in the morning or right after a major food source has been discovered. The vibration frequency is around 15–20 Hz – enough to get attention and get bees going. 5. The Stop Signal Now, this one's all about inhibition. Let's say a scout returns and starts doing a waggle dance for a site she found – but that site is dangerous. Maybe there are predators or aggressive competitors. Another bee who visited that same site comes over and delivers a stop signal. She literally head-butts the dancer and emits a short "toot" – a sharp vibration. Often, this causes the dancing bee to stop or reduce her dancing. It's an elegant way the colony uses counter-communication to avoid wasting time – or worse, losing bees to a bad decision. 6. Buzzing or Piping (During Swarming) When a colony is preparing to swarm, communication ramps up – and one signal is buzzing or piping. Scout bees that find potential nest sites will return and produce vibratory signals – often involving wing buzzing and abdomen vibrations – to stimulate other scouts or prepare the queen for flight. These signals help coordinate the swarm, so the entire group can move together when it's time. 7. Grooming Invitation Last but not least, we have the grooming invitation dance. This one is about it's about hygiene and colony health. Bees can't reach every part of their own bodies, especially when dealing with Varroa mites or debris. So, they ask for help. The signaling bee raises her abdomen, sometimes vibrates or shudders slightly, spreads her legs – especially the forelegs – and may wave her antennae. She often rotates or pivots slightly but remains stationary, essentially saying: "Please groom me." Research by Land and Seeley (2004) suggests that there might be low-level vibrational cues transmitted through the comb or direct contact – not as pronounced as in other dances, but still communicative. This dance plays a vital role in the colony's social immunity – helping bees remove pests and maintain overall health. This behavior is called "allogrooming," an aspect of social immunity when bees groom each other. To truly understand how bees interpret vibrational messages, we need to explore the sensory systems that detect them – starting with "chordotonal organs." These are specialized internal sensory structures – essentially stretch and vibration receptors – that detect mechanical changes inside the body. They're a type of sensory organ that allows bees to perceive movements and

vibrations both in their environment and within their own bodies. Honeybees have two main chordotonal organs that are particularly important for communication and navigation. Together, they allow the bee to detect both airborne vibrations, like the wingbeats of other bees, and substrate-borne vibrations, like the subtle tremors traveling through the hive. By interpreting these signals, bees can coordinate complex behaviors, from foraging to the waggle dance, and respond to their environment with incredible precision. One of these chordotonal organs is the Johnston's organ, or JO. The JO is located in the second segment of the antenna, called the pedicel. It's responsible for detecting deflections of the flagellum, which is the long, highly mobile part of the antenna made up of many small segments called flagellomeres. This mobility and sensitivity make the flagellum crucial for a wide range of functions: detecting airflow, sensing odors and pheromones, and helping bees navigate, forage, and communicate. One of the interesting things about the JO is how it interprets airborne vibrations. This includes both environmental signals, like wind, and near-field communication, like the wingbeats of other bees during the waggle dance (the term "near-field" implies that the frequency of wingbeats can only be detected by a follower in close proximity - in fact, a follower needs to be within 5 cm to accurately detect the dancer's wingbeats). It's important to note that the JO isn't directly activated by air currents. Instead, it responds to the resonance of flagellar vibrations caused by tiny air-particle movements (think how a wine glass shatters at a resonant frequency). In honeybees, the pedicel acts like a pivot: even a slight displacement at the tip of the flagellum creates distortions at the base, which the JO can detect. Structurally, the JO is composed of hundreds of scolopale cells inside the pedicel that sense these distortions between the pedicel and the flagellum. Around 240 scolopales are attached around the base of the flagellum through specialized attachment cells in the pedicel, allowing the JO to detect vibrations from all directions around the antenna's central axis. Zooming in further, the JO has 300–320 sensory units, or scolopidia, which are connected to 48 knobs at the antenna base. Each unit contains neurons with cilia (tiny, hair-like projections on the neurons within each sensory unit) that respond to vibrations. This combination of structural and mechanical specialization allows the antenna to be incredibly sensitive to low-intensity signals within the 265–350 Hz range, while filtering out excessively strong signals. This fine-tuning is exactly what enables bees to detect subtle wingbeat vibrations during the waggle dance, providing precise information for navigation and communication. Bees don't rely solely on airborne sounds to communicate – they also pick up vibrations traveling through the surfaces they're standing on, like the comb or the hive itself. To do this, they have specialized chordotonal organs in their legs called subgenual organs, which are located in the tibia just below the knee joint. Each subgenual organ is essentially a cluster of sensory neurons suspended in hemolymph, the bee's circulatory fluid. This fluid-filled setup is crucial: it allows the organ to pick up low-frequency substrate-borne vibrations, typically in the range of 10 to 1000 Hz. When the surface the bee is standing on vibrates – say, from the movements of other bees or the waggle dance – these vibrations are transmitted through the leg and into the subgenual organ. The organ converts these mechanical vibrations into nerve signals that are sent to the bee's brain. This lets the bee detect subtle vibrations traveling through the comb, providing another layer of communication beyond airborne sounds. Essentially, the subgenual organ acts as a highly sensitive "vibration detector," tuned to pick up the low-frequency signals that are critical for social coordination in the hive. In addition to the chordotonal organs, honeybees are covered with sensilla. A sensillum is a tiny sensory organ found all over a bee's body. Many of them look like small hairs or bristles, but not all sensilla are hair-like. Some are peg-shaped, plate-like, or dome-shaped, depending on what they do. These structures are packed with sensory neurons and serve as the bee's "sensing stations" for the outside world. Each sensillum is specialized – some respond to physical pressure or movement, others to chemical cues like pheromones or floral scents. Sensilla allow bees to detect: - Touch and vibration (mechanosensory sensilla) - Odors (olfactory sensilla) - Taste (gustatory sensilla) - Humidity and temperature - CO₂ levels (in some cases) Many of them are the hairs on a bee's body – when you see those tiny hairs on a bee's antennae, legs, or thorax, a lot of them are mechanosensory or chemosensory sensilla. On the antennae, sensilla work with the Johnston's organ

to detect airborne vibrations (e.g. wingbeats, piping). On the legs, they work with the subgenual organ to detect vibrations through the comb (e.g. waggle or tremble dances). On the body, they detect direct contact, like head-butting in a stop signal or tactile grooming invites. So what makes a bee buzz? A honeybee's thoracic muscles can flex and contract at a frequency of 200 to 250 times per second to create vibrations. The bee's thorax contains two main sets of flight muscles that work against each other, hence they are called "antagonistic" or "asynchronous" muscles: - Dorso-ventral (DV) muscles: These run from the top (tergum) to the bottom (sternum) of the thorax. - Dorsal-longitudinal (DL): These run lengthwise in the thorax. The defining feature of asynchronous muscles is their "stretch activation" mechanism. A single nerve impulse to one set of muscles starts an oscillating cycle: When the DV muscles contract, they deform the thorax and stretch the DL muscles. The stretching triggers the DL muscles to contract automatically, pulling the thorax in the opposite direction and stretching the DV muscles. This rapid, alternating contraction cycle continues at a high frequency, causing the entire thorax to vibrate. Bees can decouple their wings from the thoracic muscles to produce vibrations without flapping their wings. Now let's put this all together by taking a deep dive into the waggle dance, which incorporates both airborne and substrate-borne vibrations. In the waggle dance, honeybee workers communicate using a sophisticated combination of body movements and subtle vibrations. Followers detect thoracic vibrations through the comb and air-jets created by the wingbeats to acquire information on the distance and direction to a food source. These two signals operate at distinct frequencies: the wingbeats vibrate at about 265 Hz, while the waggle movements occur at 12–15 Hz. The wingbeats create weak air vibrations strongest just behind the dancer's abdomen, where followers crowd close to detect using their antennae. The angle of the waggle run indicates the direction to fly relative to the sun, while the duration of the waggle vibration encodes distance. Interestingly, distance isn't measured strictly geographically: as Karl von Frisch noted in "The Dancing Bees: An Account of the Life and Senses of the Honeybee," if a bee encounters a strong headwind or must fly uphill, its calculation of distance depends on the effort and time required. A research paper "Dynamic antennal positioning, etc." in 2024 proposes (and tests) that followers don't align themselves to the angle of the dancer but follow at various angles and use antenna position and their own sense of gravity (i.e. which way is "down") to interpret the dance direction correctly from different viewing angles. The followers deflect their antennae depending where they are in relation to the dancer and, knowing how their own head is oriented, compute this antennal angle in relation to gravity to determine direction and distance. They also propose a neural (brain) circuit model showing how that antennal input might get combined internally to produce the correct "vector" (direction to fly) reading. Fascinatingly, the resonance frequency of the flagellum is often tuned to match the frequency of the dancer's wingbeats – in the range of 265–350 Hz range. Only small changes in vibration are needed to be detected by the JO, and if the vibration becomes too strong, the antenna temporarily stops responding. It has been suggested that this mechanical property helps prevent the sensory neurons of the JO from becoming overstimulated. This acts as a natural filter to prevent sensory overload, especially when multiple dances occur nearby. Inside the bee's brain, vibration signals from the antenna are processed along two main pathways. One pathway links vibration information with motion and visual processing, while the other integrates vibration with body-angle cues. This dual processing allows follower bees to accurately extract both the direction and distance from the waggle dance. For the dance to faithfully represent distance, the dancing bee must measure her own flight precisely, which requires fine-tuned control of speed and motion. The JO plays a crucial role here, detecting flight velocity so that bees can gauge how fast they're moving. Signals from the JO are processed in brain regions like the PPL (posterior protocerebral lobe) and a region called the DL-dSEG (dorsal lobe–dorsal subesophageal ganglion complex). These regions integrate vibrations and body movement sensations and visual cues to regulate flight speed, ensuring that the dance accurately translates real-world distance into waggle duration and intensity. In addition to vibrations, touch is also essential: followers use their antennae to track the dancer's body angle. Bees preferentially follow dances on open comb cells rather than capped cells, likely because open surfaces transmit fine-scale signals

more effectively. Complementing the waggle dance, bees often engage in trophallaxis – sharing nectar before or after dancing – allowing followers to learn not only where to forage but also the taste and scent profile of the flowers. Ultimately, the waggle dance works because of a tight integration of vibration, tactile cues, and shared context. Every level, from sensory detection to neural processing to social learning, is finely tuned to convey critical information. These same mechanisms are used by scout bees during swarming to communicate potential nest sites, demonstrating how versatile and sophisticated the waggle dance truly is. In the world of bees, sound is vibration. That’s the core of their communication system – everything they ‘say’ is felt or sensed, not spoken. Bees use both airborne and substrate-borne vibrations – signals that travel through the air, like wingbeats and buzzing, and signals that move through the hive itself, like the comb shaking during a waggle dance. They’ve got specialized tools for both: The Johnston’s organ in their antennae picks up airborne signals. The subgenual organ in their legs detects vibrations through surfaces – like wax or wood. And externally honeybee’s are covered with sensillum – tiny hairs or bristles packed with neurons to detect sound and vibration. Their communication toolkit includes wingbeats, thoracic muscle vibrations, and body movements, all encoded with specific frequencies and rhythms. And it’s not just vibration – they also rely on touch, taste, and smell to make sense of the world around them. Big picture? It’s a complex, multi-sensory, vibration-based language that keeps the entire colony in sync and thriving.

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